Jack Do

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Education

Duke University - *Sophomore* *Sep 2024 – May 2028*
B.S.E. Mechanical Engineering & Aerospace Certificate; B.Sc. Computer Science; 3.97 GPA

Courses: Thermodynamics, Dynamics, Junior Mechanical Design, Computer Architecture, Advanced Web Dev, Constructal Theory.

Experience

Aerodynamics Simulation Researcher - *Duke Aeroelasticity Group* *Sep 2025 - Present*

- Perform coupled FSI simulations with ANSYS Structural & Fluent on Duke Compute Cluster for NASA HyMAX Project.

Shop Hand & Teaching Assistant - *Duke Pratt Student Machine Shop* *Sep 2025 - Present*

- Support machining classwork, junior & senior design projects by providing machining expertise and shop safety oversight.

Avionics & Separation Lead Engineer, Liquid Propulsion Engineer - *Duke AERO* *Jun 2025 - Present*

- Direct and oversee Avionics & Separation system development for Duke AERO's 2025-2026 design cycle.
- Re-engineer tri-blade airbrakes control system for 30% weight reduction and optimized blade profile for maximum apogee targeting.
- Lead redesign of canards system for 35% increase in control authority and 45% reduction in manufacturing & integration time.
- Design first-iteration comprehensive liquid engine flight stack architecture, support efforts leading up to static hotfire of ENO.

Avionics, Structures, Solid & Liquid Propulsion Design Engineer - *Duke AERO* *Sep 2024 - Jun 2025*

- Led FINSight V2 - innovative dual-layer fin PCB with embedded accelerometers & DAQ system to characterize rocket fin flutter.
- In charge of design & analysis of IP-67 rated, shock and blast-proof black box housing & PCB, validated with ANSYS FEA.
- Researched & analyzed canard-tail fin aerodynamic coupling & control surface flutter for canard roll-stabilization system. Developed closed-loop PI ramping control process. Selected for podium presentation at IREC 2025 as an outstanding project.
- Designed and optimized Solid Propulsion's static fire test stand to withstand over 35000Ns of impulse.
- Performed ANSYS FEA analyses on aluminum thrust plate withstanding over 5000N of thrust & fiberglass nosecone assembly
- Machined 6" Aluminum motor closures to ±.002" tolerance, manufactured CFRP airbrake blades & fiberglass body tubes.
- Conducted NOX & Kerosene injector simulations on ANSYS Fluent to validate injection pattern and pressure losses.
- Developed NOX & Kerosene tank weighing stand for live data logging & monitoring during cold flow & hot fire test.
- Assisted with development of LABView-based data logging & control system for Liquid Propulsion's engine, ENO.

Mechanical Engineering Intern - *GEA Group Vietnam - Liquid & Powder Technology Team* *Apr 2024 - Aug 2024*

- Designed & documented basic P&IDs; conducted on-site tank, piping & valves inspections & maintenance for F&B factories.
- Developed general EXCEL calculating tools for GEA's homogenizer flow & heat exchange systems.
- Projects involved: Lothamilk Yogurt Production Line Expansion & Ba Vi Milk Production Line Upgrade

Robotics Research Assistant Intern - *Chirikjian Lab, National University of Singapore* *Mar 2023 - Jan 2024*

- Assisted 2 grad students' research on Robot Object Affordance Imagination for motion planning with probabilistic methods.
- In charge of Gaussian filter parameter tuning for Computer Vision collision detection, streamlined tuning process by 23%.
- Assisted with post-publication paper reviews on Open Containability and Sitting Affordance via physical simulations.
- Performed 7-DOF robot arm pick & pour sims with MATLAB, ROS-2 and PyBullet, upgraded 1500+ lines codebase to Python3.

University Makerspace Technician - *Maker Innovation Space, VN-UK University* *Dec 2022, Jun 2023, Dec 2023*

- Assisted 50+ university student with course projects, ran workshops on 3D Printing & Laser CNC for high school students.
- Designed 50×50 cm^2 wind tunnel, visualized flow separation & measured different NACA airfoils' L-D performance.

Startup Project - *PrintPulse3D: 3D Printer Custom Build & Maintenance Service* *Oct 2022 - Dec 2023*

- Specialized as one of Singapore's largest 3D printer maintenance services; assembled & serviced 60+ printers over 12 months.
- Provided client consultation services on Additive Manufacturing design & production techniques for optimization at scale.

Technologies

Relevant Skills: ANSYS Structural & Fluent, GD&T, Tolerance Stackup, Precision Manual Machining, Composite Layup & Molding, CAD/CAM (SolidWorks, Onshape, AutoCAD, Fusion360), PCB design (Altium, KiCAD, Eagle), DFM & AM Optimization, Aircraft Design Suite (Athena Vortex Lattice, XFOIL & XFLR5, OpenVSP, MachUp)

Programming, Control Languages & Frameworks: Python, C, C++, Java, MATLAB, MIPS Assembly, HTML, CSS, JS, TypeScript, Vue, React, Docker, ROS, LABView & NI-DAQmx